

CINE-SYNAPSE / Studio OS

End-to-End Manual Testing & Quality Assurance Guide

System Release: 2.4.0-Production | **Active Benchmarking:** The Matrix (4 Fight Clips)
Industry Standards: OMC ISO-1004, TPN Gold Shield v5.2, C2PA v2.1, SAG-AFTRA 2026 Theatrical Agreement

Target Host: http://localhost:5173	API Engine: FastAPI (Port 8000)	Database: ClickHouse OLAP
Active Studio: Silver Pictures / Village Roadshow	Active Project: The Matrix (MATRIX-2026)	Storage Capacity: 25.0 PB Multi-Tier

Executive Overview & Test Scope

This verification manual provides a rigorous, button-by-button test procedure for CINE-SYNAPSE Studio OS. Every single interactive button, modal window, drawer, slider, and workflow developed across all 9 screens, the global header, and the real-time status bar is systematically tested from a clean scratch slate using **4 fight clips from The Matrix**.

Phase 0: Clean Slate Tenant & Project Initialization

Test Case 0.1: Register New Studio Organization (Tenant)

- **Action:** Locate the Studio Tenant Selector in the top header. Click **Register New Studio...**
- **Input Values:** Studio Name = Silver Pictures / Village Roadshow, Slug = silver_pictures, Short Code = SILV, Icon = **■■■**.
- **Verification:** Click *Cancel* to verify clean dismissal, re-open, click *Register Studio*, verify green confirmation toast and header dropdown update.

Test Case 0.2: Create Isolated Movie Project from Scratch

- **Action:** In the top header, click the **New Project** button.
- **Input Values:** Title = The Matrix: Resurrections & Revolutions, Code = MATRIX-2026, Director = Lana & Lilly Wachowski, FPS = 24.000 fps, Aspect Ratio = 2.39:1 Scope.
- **Template Mode:** Select **Clean Blank Slate** (0 scenes, 0 takes, 0 likeness riders).
- **Verification:** Click *Create Movie Project*. Verify project dropdown selects the new Matrix project and all views initialize in an isolated namespace.

Phase 1: Global Header, Status Bar & System Overlays

Test Case 1.1: Header Controls, Quick Launchers & Scenario Bar

- **Studio & Project Switchers:** Switch between tenants. Verify active project list cascades immediately without cross-tenant bleed.
- **Storage Vault Button:** Click to verify direct transition to Screen 9 Storage Vault.
- **Logs Button:** Click to open the bottom SRE Telemetry & Audit Logger drawer.
- **Scenarios Menu:** Test all 4 hackathon triggers: 1. *Trigger 23.976 Drift*, 2. *Breach Meal Penalty*, 3. *Synthesize Past SAG Cap*, 4. *Dispatch ShotGrid Inpaint*. Verify immediate state mutation toasts.
- **User Profile Avatar:** Inspect active user persona, assigned role, and tenant session ID.

Test Case 1.2: Persistent Bottom Status Bar (StatusBar.tsx)

- **SLA & Telemetry:** Verify ClickHouse: 11.4ms, Gemini 1.5 Pro | 1.2M ctx, C2PA: Root-of-Trust Valid, OMC ISO-1004 / TPN Gold.
- **High-Speed Throughput:** Verify live network meter (800G RoCE: 3.8 GB/s) and active camera slate indicator.
- **Dock Buttons:** Click [■ Vault] and [■ SRE Logs] shortcuts in the status bar to verify responsive launch.

Phase 2: Screen 1 — Autonomous Production Graph

Test Case 2.1: Graph Navigation & Node Dependency Inspection

- **KPI Filters:** Click All Nodes, Production Pipeline, Sentry & Security, Distribution & Legal.
- **Node Clicking:** Click on On-Set Camera Sentry, SAG Likeness Ledger, and Storage Vault. Verify glowing selection border and side-panel inspection details.

Test Case 2.2: 5-Dimensional Autonomous Consensus Check

- **Action:** Click [■ Run 5-D Consensus Check].
- **Simulation Progression:** Verify step-by-step telemetry: ClickHouse Graph query -> SAG Schedule A cap check -> Spherex 190-territory scan -> H100 GPU allocation.
- **Verdict Card:** Verify APPROVED_WITH_AUTOMATED_REMEDIATION, ShotGrid ticket generation (SG-TASK-8492), and 0.1% audio pull-up remediation.

Phase 3: Screen 2 — On-Set Camera Sentry (Matrix Clips Testing)

This is the primary production screen where the user tests their 4 Matrix fight clips in multi-camera sync.

Test Case 3.1: Mount 4 Matrix Fight Clips to Camera Feeds

- **Camera A (ARRI ALEXA 35):** Drag & drop **Clip 1 (Dojo Sparring: Neo vs Morpheus)** onto Camera A card.
- **Camera B (RED V-RAPTOR XL):** Drag & drop **Clip 2 (Subway Fight vs Agent Smith)** onto Camera B card.
- **Camera C (Sony VENICE 2):** Drag & drop **Clip 3 (Rooftop Bullet-Time Acrobatics)** onto Camera C card.
- **Camera D (Aux Witness):** Drag & drop **Clip 4 (Burly Brawl: 100-Agent Smith Clones)** onto Camera D card.
- **Verification:** Verify video players load, source switches to file, and resolution/framerate display on HUD.

Test Case 3.2: Master Transport & Playback Engine

- **Buttons:** Click [■ PLAY (Spacebar), ■ PAUSE (Spacebar), Step +1 Frame (L), Step -1 Frame (J)].
- **Timecode Sync:** Drag SMPTE scrubbing bar; verify synchronized scrubbing across all 4 clips.
- **Sub-Clipping:** Press [to Mark IN, press] to Mark OUT on the combat impact moment.

Test Case 3.3: DIT Assist Toolbelt (False Color, Peaking, Scopes, CDL)

- **False Color:** Toggle on; verify ARRI LogC4 thermal luminance overlay over Matrix martial arts footage.
- **Focus Peaking:** Toggle on; verify green edge highlight illuminates punch and kick contact points.
- **Zebra Stripes:** Toggle on; verify 95% IRE highlight zebra hatching appears over bright studio lights.
- **Frame Guides & De-Squeeze:** Test 2.39:1 Scope, 1.85:1 Flat, 1.43:1 IMAX, and 2.0x Anamorphic unsqueeze.
- **Real-Time Scopes (W):** Inspect Waveform monitor, Vectorscope (Neo skin tone alignment), and RGB Parade.
- **ASC-CDL Grading:** Adjust Slope, Offset, Power, and Saturation to dial in the iconic Matrix green tint.

Test Case 3.4: Live Slate Setup & Take Verdicts

- **Action:** Click [+ New Slate / Setup]. Enter Scene: 102, Setup: B, Take: 1, Lens: Cooke 40mm T2.0. Click *Apply Slate*.
- **■ CIRCLE TAKE:** Click to mark preferred editorial take; verify conform log written to ClickHouse.
- **■ HOLD TAKE:** Click to mark backup take.
- **■ NG (NO GOOD):** Click to reject take with failure reason code (e.g. Focus buzz or Drop frame).

Test Case 3.5: Ingest Matrix & Sentry Drift Remediation

- **Ingest Matrix Tab:** Review cross-camera genlock lock status, PTP Grandmaster Clock, and 10GbE bitrates.
- **Auto-Remediate:** On Camera B drift alert (+1.5 frames), click [■ Auto-Remediate]. Verify timecode pull-up filter resolves breach.

Phase 4: Screen 3 — SAG-AFTRA Digital Replica Likeness Ledger

Test Case 4.1: Register Talent Riders (Keanu Reeves & Hugo Weaving)

- **Action:** In Screen 3, click **Add Performer Rider**.
- **Keanu Reeves:** Role = Neo / Thomas Anderson, Contract = SAG-2026-KR-NEO-001, Schedule = SCHEDULE_F, Authorized = 180.0s, Residual = \$500/s.
- **Hugo Weaving:** Role = Agent Smith, Contract = SAG-2026-HW-SMITH-002, Schedule = SCHEDULE_A, Authorized = 60.0s, Residual = \$350/s.
- **Verification:** Verify performer cards render with accurate quota bars, contract IDs, and initial 0.0s burn rate.

Test Case 4.2: Top-Up Quota (+15s Extension)

- **Action:** On Keanu Reeves' card, click the **+15s** button.
- **Verification:** Toast confirms NO FAKES Act amendment signed; authorized cap increments to 195.0s.

Test Case 4.3: Synthetic Double Inspector (Gaussian Splatting & A/B Wipe)

- **Action:** On Keanu Reeves' card, click **Double**.
- **Inspection Modes:** Test **A/B Split Wipe** (drag slider 0-100%), **Photogrammetry Mesh**, **Neural Composite**, and **Landmark Delta**.
- **HUD Metrics:** Verify Scan Fidelity 99.6%, Landmark Deviation 0.038 mm, and ACEScg color gamut match.

Test Case 4.4: C2PA Cryptographic Provenance Manifest Viewer

- **Action:** Click the **c2pa:sha256:...** link on any performer card.
- **Verification:** Verify DigiCert Studio CA root-of-trust, ARRI hardware enclave seal, and click **Copy SHA-256 Hash**.

Test Case 4.5: Performer Audit Drawer & Live Shot Deduction Simulator

- **Action:** On Hugo Weaving's card, click **Audit**.
- **Permitted vs Prohibited Matrix:** Test clauses: *Dangerous wirework* (Permitted) vs *Dialogue synth* (Prohibited Section 4(B)).
- **Deduction Simulator:** Enter Scene: Scene 104, Take: 3, Seconds: 12.5s, Desc: Courtyard multi-Smith clone. Click *Record Live Shot Deduction*.
- **Verification:** Verify Hugo Weaving's used seconds increment by 12.5s and accrued residuals update immediately.

Test Case 4.6: Export SAG Section 43 Compliance Packet

- **Action:** Inside Audit Drawer, click **Export Section 43 Audit Packet**.
- **Buttons:** Test **Copy JSON Payload**, **Download Formal JSON Packet**, and **Print Formal Delivery Packet**.

Phase 5: Screen 4 — 190-Territory Distribution Compliance & Inpaint Hub

Test Case 5.1: Territory Inspection & Infraction Rules

- **Territory Grid:** Click **Saudi Arabia (GCAM)**, **China (NRTA)**, **Singapore (IMDA)**, and **United States (MPAA)**.
- **Infraction Cards:** Inspect cultural rules regarding alcohol, violent blood splatter, and supernatural content.

Test Case 5.2: Generative Inpaint A/B Comparison & ShotGrid Dispatch

- **Action:** Click **Toggle Inpaint Preview**. Observe side-by-side comparison of original plate vs AI-inpainted clean plate.
- **ShotGrid Dispatch:** Click **Dispatch to ShotGrid / Autodesk Flow**. Verify ticket SG-TASK-8492 generated and territory status transitions to CLEARED.

Phase 6: Screen 5 — SaaS Tenant Portal & IAM Security

Test Case 6.1: Personas & Role-Based Access Control (RBAC)

- **Persona Switching:** Switch between *Director*, *Production Attorney (Legal)*, *DIT / Color Engineer*, *VFX Supervisor*, and *Cloud SRE*.

Test Case 6.2: Single Sign-On (SSO) Handshake Simulation

- **Providers:** Click Okta Enterprise SSO, Ping Identity, and Google Workspace SAML 2.0. Verify JWT token issuance.

Test Case 6.3: RLS Policy Explorer & Security Audit

- **RLS Table:** Inspect ClickHouse and PostgreSQL row-level security permissions per role.
- **Session Audit:** Test Revoke Token / Sign Out (status = REVOKED), re-authenticate, and click Proceed to Studio OS Dashboard.

Phase 7: Screen 6 — Timeline Ingest & Conform Hub

Test Case 7.1: Timeline File Drag & Drop Ingestion

- **Dropzone:** Drag & drop an OpenTimelineIO (.otio) or EDL file, or click **[■ Load Sample Matrix OTIO Timeline]**.

Test Case 7.2: Multi-Track Visualizer & Clip Inspector

- **Timeline Tracks:** Review V1 (Master RAW Plates), V2 (Neural Inpaint Pass), and A1-A3 (Production Audio).
- **Clip Inspector:** Click clip segments; inspect lens metadata, timecode IN/OUT, C2PA seal, and click Conform & Dispatch to VFX Pipeline.

Phase 8: Screen 7 — Studio Copilot & Professional Side-Tools

Test Case 8.1: Gemini 1.5 Pro Studio Copilot

- **AI Prompting:** Enter: 'Analyze timecode drift between Cam A and Cam B in Scene 102 and check Keanu Reeves SAG quota.'
- **ClickHouse SQL:** Verify copilot returns synthesis accompanied by executable SQL query on the production graph.

Test Case 8.2: SMPTE Voice Scratchpad (Audio Memo Dictation)

- **Dictation:** Click **[■ Start Audio Memo Dictation]**, latch live SMPTE timecode, enter note, assign department (VFX), click Save Voice Memo.

Test Case 8.3: Look Library A/B Split Comparator & 3D LUT Export

- **Split Slider:** Test WIPE, DIFFERENCE, and FALSE_COLOR modes comparing LogC4 plate against Matrix Green LUT.
- **LUT Export:** Click **[■ Export 33-point .CUBE 3D LUT]** to download LUT, then click **[■ Dispatch LUT to DIT Cart]**.

Phase 9: Screen 8 — Studio War Room & Multi-Party Collaboration

Test Case 9.1: Live Video Huddle Simulation

- **Video Grid:** Verify 5-way video grid (Director, VFX Sup, DIT, Legal, Camera Op) with active speaker indicators.
- **Controls:** Click Mute Mic, Camera Toggle, and Share Screen buttons.

Test Case 9.2: Channel Messaging & SMPTE Timecode Latch

- **Channels:** Switch between #on-set-camera-comms, #vfx-generative-pipeline, #legal-sag-approvals.

- **Message:** Type take note with SMPTE timecode latch toggle enabled, click Send, verify instant delivery.

Test Case 9.3: Gemini Live Transcription & Action Items

- **Review:** Review live speech-to-text rolling feed and AI-extracted action items. Click **[■ Push Action Items to ShotGrid / Jira]**.

Test Case 9.4: NexGuard Invisible Forensic Watermarking

- **Inspect & Leak Test:** Click Inspect Forensic Payload, review TPN+ Level 3 hash, click Run Leak Simulation to verify bit-exact leak trace.

Test Case 9.5: Call Sheet & Union Labor Logistics Mode

- **Toggle:** Switch mode to Call Sheet & Logistics. Review crew calls, meal penalty countdown, and 12-hour turnaround tracker.

Phase 10: Screen 9 — Cloud Storage & Security Vault

Test Case 10.1: Multi-Petabyte Tier Architecture Dashboard

- **Storage Tiers Tab:** Inspect Tier 0 (Hot NVMe 2.5 PB @ 800 Gbps RoCE v2), Tier 1 (Warm GCS 7.5 PB), Tier 2 (Cold WORM Tape 15.0 PB).
- **Capacity Gauges:** Verify total 25.0 PB capacity display and auto-tiering rules.

Test Case 10.2: Camera-to-Cloud (C2C) Real-Time Ingress Monitor

- **C2C Ingress Tab:** Inspect Cam A (1.2 GB/s), Cam B (1.4 GB/s), Cam C (1.2 GB/s) streaming pipes.
- **Proxy Transcode Matrix:** Verify automated background transcode to Apple ProRes 422 Proxy and H.265 streaming dailies.

Test Case 10.3: TPN Gold Shield & Cloud KMS BYOK Center

- **Security & KMS Tab:** Review TPN Gold Shield v5.2 compliance posture and FIPS 140-3 HSM BYOK master key ID.
- **[■ Rotate KMS Key Now]:** Click button; verify zero-downtime key rotation toast (rotates to v4) and timestamp update.
- **[■■ Run SHA-256 Parity & Integrity Audit]:** Click button; verify progress bar and audit confirmation (14,892 C2PA hashes verified).

Test Case 10.4: Cryptographic Encrypted Asset Explorer

- **Asset Explorer Tab:** Search MATRIX or ari or RAW; verify dynamic asset table filtering.
- **Inspect C2PA:** Click button on any asset row; verify C2PA cryptographic manifest modal opens with signed hardware certificates.

Appendix: Matrix Fight Clips Mapping & Test Case Matrix

Clip Name	Narrative Sequence	Recommended Camera	Primary Test Scenarios Exercised
Clip 1	Dojo Sparring (Neo vs Morpheus)	Camera A (ARRI ALEXA 35)	Master Transport (J/K/L), False Color exposure, Zebra highlights, ASC-CDL Green Matrix Tint, Focus Peaking on martial arts impacts.
Clip 2	Subway Station Fight (Neo vs Agent Smith)	Camera B (RED V-RAPTOR XL)	Anamorphic 2.39:1 Scope Framing, Genlock Ingest Matrix, Timecode Drift (+1.5 frames) auto-remediation via audio pull-up.
Clip 3	Rooftop Bullet-Time (High-Speed Acrobatics)	Camera C (Sony VENICE 2)	High-Speed 120 fps capture, Video Scopes (Waveform & Vectorscope skin tone line), In/Out Marker sub-clipping.
Clip 4	Burly Brawl (100-Agent Smith Clones)	Camera D (Aux / VFX Plate)	SAG Likeness Ledger deduction (12.5s for Hugo Weaving), Synthetic Double A/B Split Wipe, C2PA Manifest audit & export.

Testing Protocol Complete. All test cases in this manual comply with the 2026 SAG-AFTRA TV/Theatrical Agreement, Federal NO FAKES Act, TPN Gold Shield v5.2, and MovieLabs 2030 Vision for Cloud-Native Production.